

Software Requirements Specification (SRS)

For Waste Food Management System

1. Introduction

1.1 Purpose

This document specifies the requirements for a Waste Food Management System designed to reduce food wastage by connecting food donors with recipients. The system allows donors to post excess food, recipients to request available food, and administrators to manage the entire platform.

1.2 Scope

The system will:

- Manage donor, recipient, and administrator accounts.
- Allow donors to add, update, view, and delete food listings.
- Allow recipients to browse available food and submit requests.
- Track food requests and their status.
- Provide administrative control over users, food listings, and requests.
- Support simple CRUD (Create, Read, Update, Delete) operations.

1.3 Definitions, Acronyms, and Abbreviations

- SRS: Software Requirements Specification
- CRUD: Create, Read, Update, Delete
- UI: User Interface
- DB: Database
- Admin: System Administrator

2. Overall Description

2.1 User Needs

Donor

- Register and log in.
- Add food details.
- Update food information.
- Delete food listings.
- View food requests.

Recipient

- Register and log in.
- Browse available food.
- Request food.
- View request status.

Administrator

- Manage users.
- Manage food listings.
- Manage food requests.
- Monitor system activities.

2.2 Assumptions and Dependencies

- Users have access to a web browser.
- Internet connectivity is available.
- The system uses a MySQL database for data storage.
- Users provide valid information during registration.

3. Functional Requirements

3.1 User Management

FR1.1

The system shall allow donors and recipients to register.

FR1.2

The system shall allow registered users to log in.

FR1.3

The system shall allow users to update their profile information.

FR1.4

The administrator shall manage user accounts.

3.2 Food Listing Management

FR2.1

Donors shall add new food listings.

FR2.2

Donors shall view their food listings.

FR2.3

Donors shall update food details.

FR2.4

Donors shall delete food listings.

3.3 Food Request Management**FR3.1**

Recipients shall browse available food.

FR3.2

Recipients shall submit food requests.

FR3.3

The system shall store request information.

FR3.4

Recipients shall view request status.

FR3.5

Donors shall view requests made for their food.

3.4 Administration**FR4.1**

Administrators shall view all users.

FR4.2

Administrators shall manage food listings.

FR4.3

Administrators shall manage food requests.

FR4.4

Administrators shall remove invalid or expired records.

3.5 Reporting

FR5.1

The system shall display the total number of food donations.

FR5.2

The system shall display food request records.

4. Non-Functional Requirements

4.1 Performance

- The system shall load pages within 3 seconds under normal conditions.
- The system shall support multiple users simultaneously.

4.2 Security

- User authentication shall be required.
- Passwords shall be stored securely.
- Only administrators shall access management functions.

4.3 Usability

- The user interface shall be simple and easy to navigate.
- The system shall provide clear messages for successful and failed operations.

4.4 Reliability

- The system shall store data accurately.
- Data shall remain available unless intentionally deleted by authorized users.

5. System Architecture

- Frontend: HTML, CSS,

- Backend: PHP
- Database: MySQL
- Architecture: Client-Server Web Application

6. Constraints

- The system shall initially support English language only.
- Users require internet access.
- The project focuses on CRUD operations and basic reporting.

7. Future Enhancements

- Email notifications for food requests.
- Real-time tracking of food collection.
- Location-based food donation services.
- Mobile application support.

8. Approval

This SRS is prepared for the development of the Waste Food Management System and is subject to review and approval before implementation.